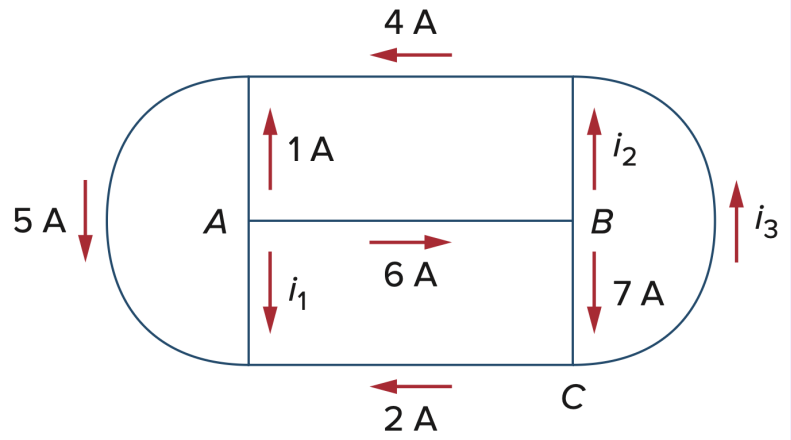
Exercise Sheet #2

DC 1

Exercise 1: KCL

Find i_1, i_2, and i_3.



Solution

node A:

$$-1\,\text{A} - 6\,\text{A} - i_1 = 0$$

$$\Rightarrow i_1 = -1\,\text{A} - 6\,\text{A} = -7\,\text{A}$$

node B:

$$6\,\text{A} - 7\,\text{A} - i_2 = 0$$

$$\Rightarrow i_2 = 6\,\text{A} - 7\,\text{A} = -1\,\text{A}$$

node C:

$$7\,\text{A} - 2\,\text{A} - i_3 = 0$$

$$\Rightarrow i_3 = 7\,\text{A} - 2\,\text{A} = 5\,\text{A}$$

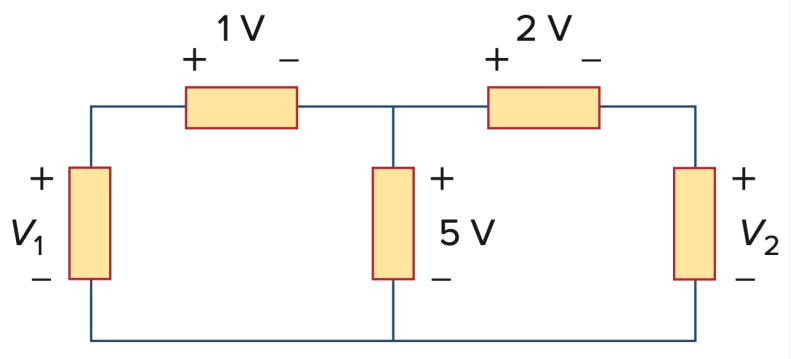
... other nodes:

$$i_2 + i_3 - 4\,\text{A} = 0$$

$$\Rightarrow i_2 = 4\,\text{A} - i_3 = -1\,\text{A}$$

Exercise 2: KVL

Calculate V_1 and V_2.



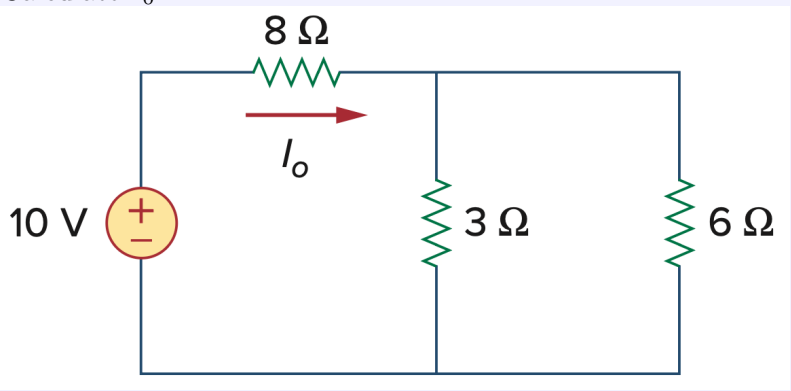
Solution

$$1\,\mathrm{V} + 5\,\mathrm{V} - V_1 = 0$$
$$\Rightarrow V_1 = 1\,\mathrm{V} + 5\,\mathrm{V} = 6\,\mathrm{V}$$

$$2\,\mathrm{V} + V_2 - 5\,\mathrm{V} = 0$$
$$\Rightarrow V_2 = 5\,\mathrm{V} - 2\,\mathrm{V} = 3\,\mathrm{V}$$

Exercise 3: Series and Parallel Resistors

Calculate I_0.



Solution

First, we calculate the equivalent resistance of the parallel resistors with $3\,\Omega$ and $6\,\Omega$, then we add the series resistor with $8\,\Omega$:

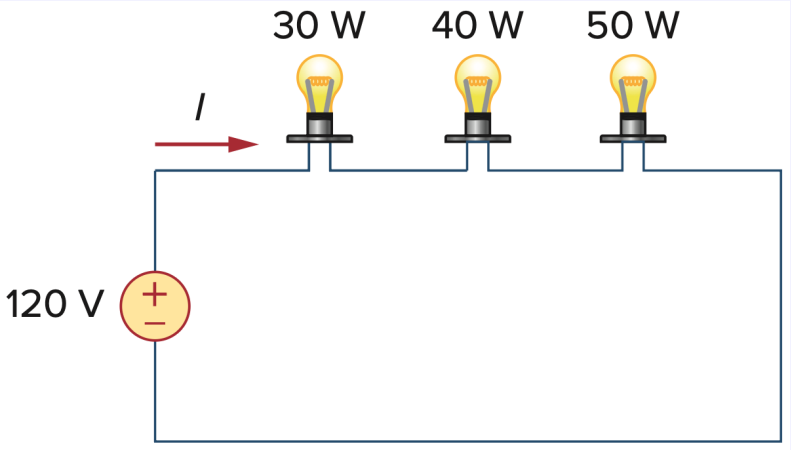
$$R = \frac{3\,\Omega \cdot 6\,\Omega}{3\,\Omega + 6\,\Omega} + 8\,\Omega = 10\,\Omega$$

Finally, we can calculate the current I_0 using Ohm's law:

$$I_o = \frac{10\,\mathrm{V}}{10\,\Omega} = 1\,\mathrm{A}$$

Exercise 4: Series vs. Parallel

Three light bulbs are connected in series to a 120 V source as shown below. Find the current I through each of the bulbs. Each bulb is rated at 120 V. How much power is each bulb absorbing? Do they generate much light?



Solution

First, we calculate the resistances of each bulb using their rated power and voltage:

$$R = \frac{V^2}{P}$$

$$R_{30\,\mathrm{W}} = \frac{(120\,\mathrm{V})^2}{30\,\mathrm{W}} = 480\,\Omega$$

$$R_{40\,\mathrm{W}} = \frac{(120\,\mathrm{V})^2}{40\,\mathrm{W}} = 360\,\Omega$$

$$R_{50\,\mathrm{W}} = \frac{(120\,\mathrm{V})^2}{50\,\mathrm{W}} = 288\,\Omega$$

Then, we calculate the equivalent resistance of the three light bulbs in series:

$$R = 480\,\Omega + 360\,\Omega + 288\,\Omega = 1128\,\Omega$$

Then, we can calculate the current I using Ohm's law:

$$I = \frac{120\,\mathrm{V}}{1128\,\Omega} = 0.1063\,\mathrm{A}$$

Finally, we can calculate the power absorbed by each bulb:

$$P = I^2 \cdot R$$

$$P_{30\,\mathrm{W}} = (0.1063\,\mathrm{A})^2 \cdot 480\,\Omega = 5.432\,\mathrm{W}$$

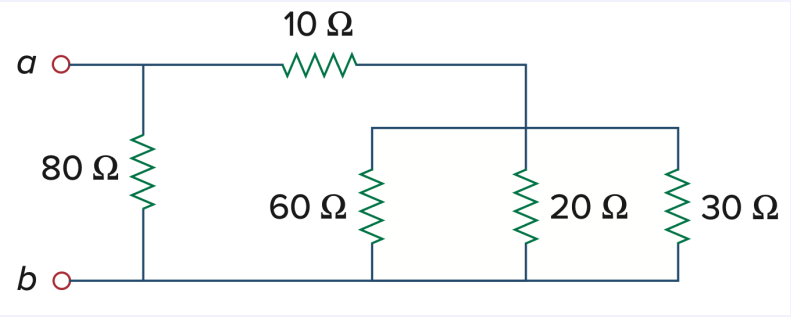
$$P_{40\,\mathrm{W}} = (0.1063\,\mathrm{A})^2 \cdot 360\,\Omega = 4.074\,\mathrm{W}$$

$$P_{50\,\mathrm{W}} = (0.1063\,\mathrm{A})^2 \cdot 288\,\Omega = 3.259\,\mathrm{W}$$

Clearly these values are well below the rated powers of each light bulb, so we would not expect very much light from any of them. To work properly, they need to be connected in parallel.

Exercise 5: Equivalent Resistance 1

Calculate the equivalent resistance R_{ab} across the terminals.



Solution

First we calculate the equivalent resistance of the parallel resistors with $60\,\Omega$, $20\,\Omega$, and $30\,\Omega$:

$$R_p = \frac{1}{\frac{1}{60\,\Omega} + \frac{1}{20\,\Omega} + \frac{1}{30\,\Omega}} = 10\,\Omega$$

Then we add the series resistor with $10\,\Omega$:

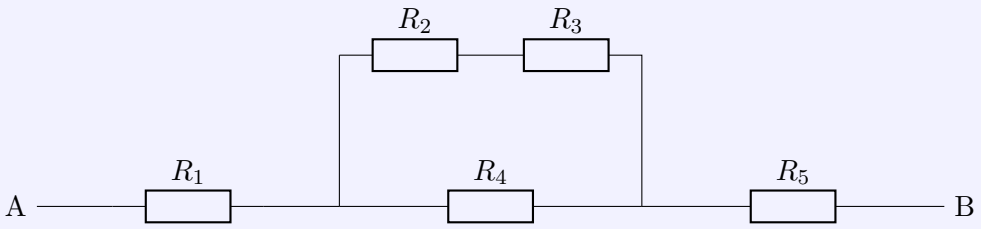
$$R_{ab} = 10\,\Omega + 10\,\Omega = 20\,\Omega$$

And finally we combine with the parallal resistor of $80\,\Omega$:

$$R_{ab} = \frac{1}{\frac{1}{20\,\Omega} + \frac{1}{80\,\Omega}} = 16\,\Omega$$

Exercise 6: Equivalent Resistance 2

Given the resistor network with $R_1 = 10\,\Omega$, $R_2 = 20\,\Omega$, $R_3 = 30\,\Omega$, $R_4 = 5\,\Omega$, $R_5 = 15\,\Omega$:



Calculate the equivalent resistance of the network between terminals A and B.

Solution

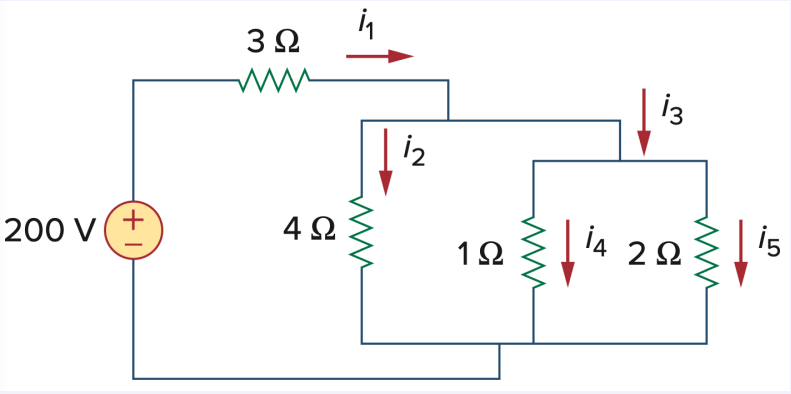
$$R_{23} = R_2 + R_3 = 20 + 30 = 50\,\Omega$$

$$\frac{1}{R_{234}} = \frac{1}{R_{23}} + \frac{1}{R_4} = \frac{1}{50} + \frac{1}{5} = \frac{11}{50}\,\frac{1}{\Omega}$$

$$R = R_1 + R_{234} + R_5 = 10 + \frac{50}{11} + 15 = 29.55\,\Omega$$

Exercise 7: Current Divider Rule

Calculate i_1 to i_5.



Solution

First, we calculate the equivalent resistance of the parallel resistors with $1\,\Omega$ and $2\,\Omega$ and call it R_3:

$$R_3 = \frac{1\,\Omega \cdot 2\,\Omega}{1\,\Omega + 2\,\Omega} = \frac{2}{3}\,\Omega$$

Then we calculate the total resistance R from the $4\,\Omega$ resistor parallel to R_3 and the $3\,\Omega$ resistor in series:

$$R = 3\,\Omega + \frac{4\,\Omega \cdot \frac{2}{3}\,\Omega}{4\,\Omega + \frac{2}{3}\,\Omega} = \frac{25}{7}\,\Omega$$

Using Ohm's law, we can calculate the total current i_1:

$$i_1 = \frac{200\,\mathrm{V}}{\frac{25}{7}\,\Omega} = 56\,\mathrm{A}$$

Then, we can calculate i_2 using current division:

$$i_2 = i_1 \cdot \frac{R_{2,3}}{R_2} = i_1 \cdot \frac{R_2 \cdot R_3}{R_2 \cdot (R_2 + R_3)} = i_1 \cdot \frac{R_3}{R_2 + R_3} = 56\,\mathrm{A} \cdot \frac{\frac{2}{3}\,\Omega}{4\,\Omega + \frac{2}{3}\,\Omega} = 8\,\mathrm{A}$$

KCL gives us i_3:

$$i_3 = i_1 - i_2 = 56\,\mathrm{A} - 8\,\mathrm{A} = 48\,\mathrm{A}$$

Again, we can calculate i_4 using current division:

$$i_4 = i_3 \cdot \frac{R_5}{R_4 + R_5} = 48\,\mathrm{A} \cdot \frac{2\,\Omega}{1\,\Omega + 2\,\Omega} = 32\,\mathrm{A}$$

Finally, KCL gives us i_5:

$$i_5 = i_3 - i_4 = 48\,\mathrm{A} - 32\,\mathrm{A} = 16\,\mathrm{A}$$

Quicker alternative:

$$v_1 = R_1 \cdot i_1 = 3\,\Omega \cdot 56\,\mathrm{A} = 168\,\mathrm{V}$$

$$v_2 = v_4 = v_5 = v - v_1 = 32\,\mathrm{V}$$

$$i_2 = \frac{v_2}{R_2} = \frac{32}{4}\,\mathrm{A} = 8\,\mathrm{A}$$

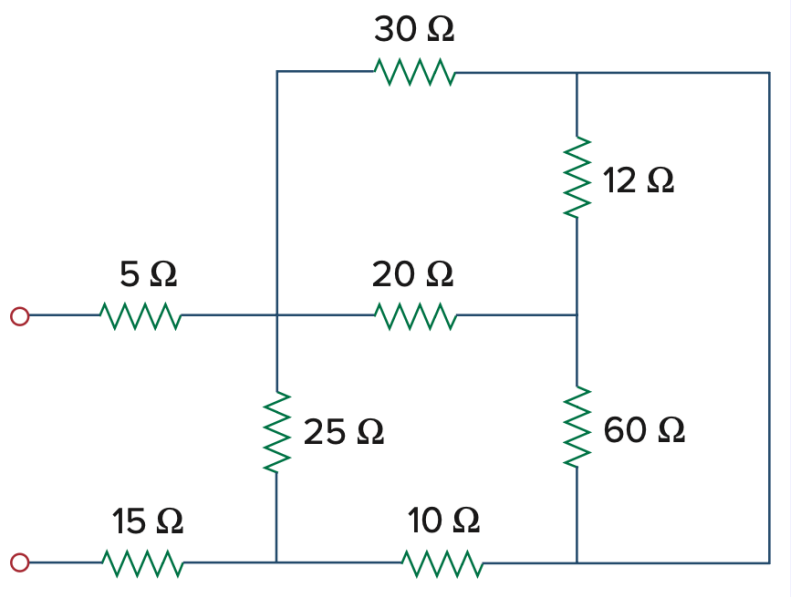
$$i_4 = \frac{v_2}{R_4} = \frac{32}{1}\,\text{A} = 32\,\text{A}$$

$$i_5 = \frac{v_2}{R_5} = \frac{32}{2}\,\text{A} = 16\,\text{A}$$

$$i_3 = i_4 + i_5 = 32\,\text{A} + 16\,\text{A} = 48\,\text{A}$$

Exercise 8: Circuit Analysis

Calculate the equivalent resistance across the terminals.



Solution

1. Combine the right vertical resistors:

$$12\,\Omega \parallel 60\,\Omega = \frac{12\,\Omega \cdot 60\,\Omega}{12\,\Omega + 60\,\Omega} = 10\,\Omega$$

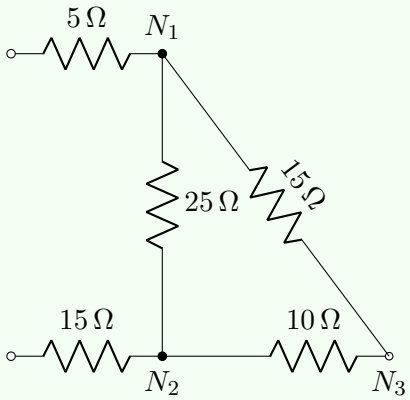
2. Series combination with the $20\,\Omega$ resistor:

$$20\,\Omega + 10\,\Omega = 30\,\Omega$$

3. Parallel combination with the other $30\,\Omega$:

$$30\,\Omega \parallel 30\,\Omega = \frac{30\,\Omega \cdot 30\,\Omega}{30\,\Omega + 30\,\Omega} = 15\,\Omega$$

After these three steps, the circuit reduces to a triangle (Δ):

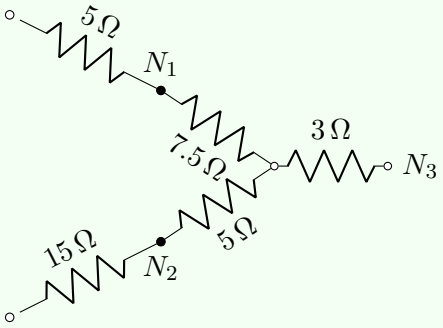


4. Convert the Δ-network to equivalent Y-resistances:

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c} = \frac{15\,\Omega \cdot 25\,\Omega}{50\,\Omega} = 7.5\,\Omega$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c} = \frac{10\,\Omega \cdot 25\,\Omega}{50\,\Omega} = 5\,\Omega$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c} = \frac{10\,\Omega \cdot 15\,\Omega}{50\,\Omega} = 3\,\Omega$$



The branch with the 3 Ω resistor connects to a dead-end node N_3, so it carries no current and can be ignored.

$$\Rightarrow R_{eq} = 5\,\Omega + 7.5\,\Omega + 5\,\Omega + 15\,\Omega = 32.5\,\Omega$$

Alternative for 4. without Δ-Y transformation:
Since N_3 is a trivial node (just one input and one equal output), you can simply calculate the series resistance of the 15 Ω and 10 Ω resistors, then combine it in parallel with the 25 Ω resistor, and finally add the series resistors with 5 Ω and 15 Ω.